SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	Sasikumar Megha	**Reg. No.:**	192471038
Section / Batch:	3rd cs&bs	**Date:**	24/07/2026

Code of Conduct

I __________________________Sasikumar Megha____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: ___Sasikumar Megha_____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
	GRAND TOTAL:	**100 Marks**	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edgecase errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors

Code Quality & Readability	20	Clean, modular, welldocumented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and welljustified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ______________ Date: __________________	Signature: ______________ Date: __________________	Signature: ______________ Date: __________________

QUESTIONS

- **Q1** Binary Search 5 marks
- **Q2** Stack 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Stack · Balancing Symbols 5 marks
- **Q6** Singly Linked List 5 marks
- **Q7** Stack 5 marks
- **Q8** Linked List 5 marks
- **Q9** Linked List 5 marks
- **Q10** Queue 5 marks

5/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
int binarysearch(int arr[],int n,int key){
    int low=0,high=n-1;
    while(low<=high){
        int mid=(low+high)/2;
            if(arr[mid]==key)
                return mid;
        else if(arr[mid]<key)
            low=mid+1;
        else
            high=mid-1;
    }
    return-1;
}
int main(){
    int arr[]={2,4,6,8,12,15};
    int n=sizeof(arr)/sizeof(arr[0]);
    int key;
    printf("enter element to search:");
    scanf("%d",&key);
    int result=binarysearch(arr,n,key);
    if(result==1)
        printf("element not found");
        else
            printf("element is found at index%d",result);
    return 0;
}
```

Test Input

```
8
```

Output

```
enter element to search:element is found at index3
```

Pending manual review

Submitted

QUESTIONS

- **Q1** Binary Search 5 marks
- **Q2** Stack 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Stack · Balancing Symbols 5 marks
- **Q6** Singly Linked List 5 marks
- **Q7** Stack 5 marks
- **Q8** Linked List 5 marks
- **Q9** Linked List 5 marks
- **Q10** Queue 5 marks

5/10 answered

Q2 **Stack** 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
int stack[5];
int top = -1;
int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }
    return stack[top--];
}

void push(int val) {
    if (top == 4) {
        printf("Overflow\n");
    } else {
        stack[++top] = val;
        printf("Pushed %d\n", val);
    }
}

int main() {
    push(10);
    push(20);
    printf("Deleted element: %d\n", pop());
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Pushed 10
Pushed 20
Deleted element: 20
```

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack · Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

5/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
int stack[5];
int top;

void init() {
    top = -1;
}

int main(void) {
    init();
    printf("Stack initialized\n");
    printf("Top = %d\n", top);
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Stack initialized
Top = -1
```

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Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != ')') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <ctype.h>
#include <string.h>

#define max 100

char stack[max];
int top = -1;

void push(char c) {
    stack[++top] = c;
}

char pop() {
    return stack[top--];
}

char peek() {
    return stack[top];
}

int precedence(char c) {
    if (c == '^') return 3;
    if (c == '*' || c == '/') return 2;
    if (c == '+' || c == '-') return 1;
    return 0;
}

void infixtopostfix(char infix[], char postfix[]) {
    int i = 0, j = 0;
    char ch;

    while ((ch = infix[i]) != '\0') {
        if (isalnum(ch)) {
            postfix[j++] = ch;
        } else if (ch == '(') {
            push(ch);
        } else if (ch == ')') {
            while (top != -1 && peek() != '(') {
                postfix[j++] = pop();
            }
            if (top != -1 && peek() == '(') {
                pop();
            }
        } else {
            while (top != -1 && precedence(peek()) >= precedence(ch)) {
                postfix[j++] = pop();
            }
            push(ch);
        }
        i++;
    }

    while (top != -1) {
        postfix[j++] = pop();
    }

    postfix[j] = '\0';
}

int main() {
    char infix[max], postfix[max];

    scanf("%99s", infix);

    infixtopostfix(infix, postfix);
    printf("%s", postfix);

    return 0;
}
```

Submitted

Test Input

```
a+b*c
```

Output

```
abc*+
```

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CO2_AT2 DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <string.h>

#define max 1000

int isbalanced(const char *expr) {
    char stack[max];
    int top = -1;

    for (int i = 0; expr[i] != '\0'; i++) {
        char ch = expr[i];

        if (ch == '(' || ch == '[' || ch == '{') {
            if (top < max - 1) {
                stack[++top] = ch;
            } else {
                return 0;
            }
        }
        else if (ch == ')' || ch == ']' || ch == '}') {
            if (top == -1) {
                return 0;
            }

            if ((ch == ')' && stack[top] == '(') ||
                (ch == ']' && stack[top] == '[') ||
                (ch == '}' && stack[top] == '{')) {
                top--;
            } else {
                return 0;
            }
        }
    }

    return (top == -1);
}

int main() {
    char expr[max];

    if (fgets(expr, sizeof(expr), stdin) == NULL) {
        return 0;
    }

    expr[strcspn(expr, "\n")] = '\0';

    if (isbalanced(expr)) {
        printf("1");
    } else {
        printf("0");
    }

    return 0;
}
```

Test Input

{[()]}

Output

1

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

🐞 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

struct node* createnode(int data) {
    struct node *newnode = (struct node*)malloc(sizeof(struct node));
    newnode->data = data;
    newnode->next = NULL;
    return newnode;
}

void append(struct node **head, int data) {
    struct node *newnode = createnode(data);

    if (*head == NULL) {
        *head = newnode;
        return;
    }

    struct node *temp = *head;
    while (temp->next != NULL) {
        temp = temp->next;
    }
    temp->next = newnode;
}

void display(struct node *head) {
    struct node *temp = head;

    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

int main() {
    struct node *head = NULL;

    append(&head, 10);
    append(&head, 20);
    append(&head, 30);
    append(&head, 40);

    printf("Linked List: ");
    display(head);

    return 0;
}
```

Submitted

Test Input

stdin input (optional)

Output

```
Linked List: 10 20 30 40
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#define max 100
int stack[max];
int top = -1;
void push(int item) {
    if (top == max - 1) {
        printf("Overflow\n");
    } else {
        stack[++top] = item;
    }
}

int pop() {
    if (top == -1) {
        printf("underflow\n");
        return -1;
    } else {
        return stack[top--];
    }
}

int main() {
    int choice, x, item;

    while (scanf("%d", &choice) == 1) {
        if (choice == 1) {
            if (scanf("%d", &x) != 1) break;
            push(x);
        } else if (choice == 2) {
            item = pop();
            if (item != -1) {
                printf("%d\n", item);
            }
        } else if (choice == 3) {
            break;
        }
    }

    return 0;
}
```

Test Input

```
1
23
1
4
```

Output

```
56
4
23
```

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Submitted

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q8 **Linked List**

5 marks

Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

struct node* createnode(int data) {
    struct node *newnode = (struct node*)malloc(sizeof(struct node));
    newnode->data = data;
    newnode->next = NULL;
    return newnode;
}

void insertend(struct node **head, int data) {
    struct node *newnode = createnode(data);

    if (*head == NULL) {
        *head = newnode;
        return;
    }

    struct node *temp = *head;
    while (temp->next != NULL) {
        temp = temp->next;
    }
    temp->next = newnode;
}

struct node* reverselist(struct node *head) {
    struct node *prev = NULL;
    struct node *current = head;
    struct node *next = NULL;

    while (current != NULL) {
        next = current->next;
        current->next = prev;
        prev = current;
        current = next;
    }

    return prev;
}

void printlist(struct node *head) {
    struct node *temp = head;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
}

void freelist(struct node *head) {
    struct node *temp;
    while (head != NULL) {
        temp = head;
        head = head->next;
        free(temp);
    }
}

int main() {
    struct node *head = NULL;
    int choice, data;

    while (scanf("%d", &choice) == 1) {
        if (choice == 1) {
            if (scanf("%d", &data) == 1) {
                insertend(&head, data);
            }
        } else if (choice == 3) {
            printlist(head);
            printf("\n");
        } else if (choice == 4) {
            head = reverselist(head);
        }
    }

    freelist(head);
    return 0;
}
```

Test Input

```
1
10
10
3
```

Output

```
10
10
10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

new.next = head

head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

struct Node *head = NULL;

void insertAtBeginning(int value) {
    struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
    if (newNode == NULL) {
        printf("Memory allocation failed\n");
        return;
    }

    newNode->data = value;
    newNode->next = head;
    head = newNode;
}

void display() {
    struct Node *temp = head;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

int main() {
    int n, value;

    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &value);
        insertAtBeginning(value);
    }

    display();
    return 0;
}
```

Test Input

```
5
10 20 30 40 50
```

Output

```
50 40 30 20 10
```

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1 Binary Search 5 marks
Q2 Stack 5 marks
Q3 Stack 5 marks
Q4 Stack 5 marks
Q5 Stack - Balancing Symbols 5 marks
Q6 Singly Linked List 5 marks
Q7 Stack 5 marks
Q8 Linked List 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

10/10 answered

Q10 **Queue** 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 100

int Q[MAX];
int front = -1, rear = -1;

int dequeue() {
    if (front == -1 || front > rear) {
        printf("Underflow\n");
        return -1;
    }

    int item = Q[front];
    front++;

    if (front > rear) {   // queue became empty
        front = rear = -1;
    }

    return item;
}

int main() {
    int n, i, x;

    scanf("%d", &n);

    if (n <= 0) {
        printf("Underflow\n");
        return 0;
    }

    front = 0;
    rear = n - 1;

    for (i = 0; i < n; i++) {
        scanf("%d", &Q[i]);
    }

    x = dequeue();

    if (x != -1) {
        printf("%d\n", x);
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

Run | Save | Submit Fix

SET 2

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q1 **Stack** 5 marks

Point out the error?

PEEK(stack):

return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int x) {
    if (top < MAX - 1) {
        stack[++top] = x;
    }
}

int peek() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }
    return stack[top];
}

int main() {
    int n, x;

    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &x);
        push(x);
    }

    x = peek();
    if (x != -1) {
        printf("%d\n", x);
    }

    return 0;
}
```

Test Input

```
5
10
20
30
```

Output

```
50
```

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- **Q1** Stack 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Doubly Linked List 5 marks
- **Q6** Queue 5 marks
- **Q7** Queue - Deque 5 marks
- **Q8** Binary Search 5 marks
- **Q9** Circular Queue 5 marks
- **Q10** Stack - Postfix Expression 5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define max 100

int queue[max];
int front = -1, rear = -1;

int isFull() {
    return ((rear + 1) % max == front);
}

int isEmpty() {
    return (front == -1);
}

void enqueue(int vehicle) {
    if (isFull()) {
        printf("Queue Full\n");
        return;
    }

    if (isEmpty()) {
        front = rear = 0;
    } else {
        rear = (rear + 1) % max;
    }

    queue[rear] = vehicle;
}

int dequeue() {
    if (isEmpty()) {
        printf("Queue Empty\n");
        return -1;
    }

    int item = queue[front];

    if (front == rear) {
        front = rear = -1;
    } else {
        front = (front + 1) % max;
    }

    return item;
}

int main() {
    int n, vehicle;

    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &vehicle);
        enqueue(vehicle);
    }

    while (!isEmpty()) {
        printf("%d ", dequeue());
    }

    return 0;
}
```

Submitted

Test Input

```
5 10 20 30 40 50
```

Output

```
10 20 30 40 50
```

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- **Q1** Stack 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Doubly Linked List 5 marks
- **Q6** Queue 5 marks
- **Q7** Queue - Deque 5 marks
- **Q8** Binary Search 5 marks
- **Q9** Circular Queue 5 marks
- **Q10** Stack - Postfix Expression 5 marks

10/10 answered

Q3 **Stack** 5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define max 100

int stack[max];
int top = -1;

void push(int value) {
    if (top >= max - 1) {
        printf("Overflow\n");
        return;
    }
    stack[++top] = value;
}

int main() {
    int n, value;

    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &value);
        push(value);
    }
    printf("stack: ");
    for(int i=top;i>=0;i--){
        printf("%d",stack[i]);
    }
printf("\n");
    return 0;
}
```

Test Input

```
5
10
20
30
```

Output

```
stack: 5040302010
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

Q4 **Stack**

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int value) {
    if (top == MAX - 1) {
        printf("Overflow\n");
        return;
    }
    stack[++top] = value;
}

int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }
    return stack[top--];
}

int main() {
    int n, value, deleted;

    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &value);
        push(value);
    }

    deleted = pop();
    if (deleted != -1) {
        printf("%d\n", deleted);
    }

    return 0;
}
```

Test Input

```
5
10
20
30
```

Output

```
50
```

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QUESTIONS

- **Q1** Stack 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Doubly Linked List 5 marks
- **Q6** Queue 5 marks
- **Q7** Queue - Deque 5 marks
- **Q8** Binary Search 5 marks
- **Q9** Circular Queue 5 marks
- **Q10** Stack - Postfix Expression 5 marks

10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

typedef struct node {
    int data;
    struct node *prev;
    struct node *next;
} node;

node *createnode(int data) {
    node *newnode = (node *)malloc(sizeof(node));
    newnode->data = data;
    newnode->prev = NULL;
    newnode->next = NULL;
    return newnode;
}

void insertend(node **head, int data) {
    node *newnode = createnode(data);

    if (*head == NULL) {
        *head = newnode;
        return;
    }

    node *temp = *head;
    while (temp->next != NULL) {
        temp = temp->next;
    }

    temp->next = newnode;
    newnode->prev = temp;
}

void deletenode(node **head, node *del) {
    if (*head == NULL || del == NULL) {
        return;
    }

    if (*head == del) {
        *head = del->next;
    }

    if (del->prev != NULL) {
        del->prev->next = del->next;
    }

    if (del->next != NULL) {
        del->next->prev = del->prev;
    }

    free(del);
}

void display(node *head) {
    node *temp = head;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

int main() {
    node *head = NULL;

    insertend(&head, 10);
    insertend(&head, 20);
    insertend(&head, 30);
    insertend(&head, 40);

    deletenode(&head, head->next);   // deletes 20

    display(head);

    return 0;
}
```

Test Input

```
4
10 20 30 40
```

Output

```
10 30 40
```

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

void enqueue(int data) {

if (rear == MAX) return;

queue[rear++] = data;

}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define max 100

int queue[max];
int front = -1, rear = -1;

void enqueue(int data) {
    if (rear == max - 1) {
        printf("Overflow\n");
        return;
    }

    if (front == -1) {
        front = 0;
    }

    queue[++rear] = data;
}

int dequeue() {
    if (front == -1 || front > rear) {
        printf("underflow\n");
        return -1;
    }

    return queue[front++];
}

int main() {
    int n, value;

    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &value);
        enqueue(value);
    }

    while (front != -1 && front <= rear) {
        printf("%d ", dequeue());
    }

    return 0;
}
```

Test Input

```
5
10
20
30
```

Output

```
10 20 30 40 50
```

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q7 **Queue - Deque**

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 100

int deque[MAX];
int front = -1, rear = -1;

void insertFront(int data) {
    if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
        printf("Overflow\n");
        return;
    }

    if (front == -1) {
        front = rear = 0;
    } else if (front == 0) {
        front = MAX - 1;
    } else {
        front = front - 1;
    }

    deque[front] = data;
}

void display() {
    if (front == -1) {
        printf("Empty\n");
        return;
    }

    int i = front;
    while (1) {
        printf("%d ", deque[i]);
        if (i == rear) break;
        i = (i + 1) % MAX;
    }
    printf("\n");
}

int main() {
    int n, value;

    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &value);
        insertFront(value);
    }

    display();
    return 0;
}
```

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Test Input

```
5
10
20
30
40
```

Output

```
50 40 30 20 10
```

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QUESTIONS

- **Q1** Stack 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Doubly Linked List 5 marks
- **Q6** Queue 5 marks
- **Q7** Queue - Deque 5 marks
- **Q8** Binary Search 5 marks
- **Q9** Circular Queue 5 marks
- **Q10** Stack - Postfix Expression 5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

int binarySearch(int arr[], int l, int r, int x) {

int mid = l + (r - l) / 2;

if (arr[mid] == x) return mid;

if (arr[mid] > x) return binarySearch(arr, l, mid, x);

return binarySearch(arr, mid, r, x);

}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int binarySearch(int arr[], int l, int r, int x) {
    if (l > r) {
        return -1;
    }

    int mid = l + (r - l) / 2;

    if (arr[mid] == x) {
        return mid;
    }

    if (arr[mid] > x) {
        return binarySearch(arr, l, mid - 1, x);
    }

    return binarySearch(arr, mid + 1, r, x);
}

int main() {
    int n, x, arr[100];

    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &x);

    int result = binarySearch(arr, 0, n - 1, x);

    printf("%d\n", result);

    return 0;
}
```

Test Input

```
5
10
20
30
```

Output

```
-1
```

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define max 100

int queue[max];
int front = -1, rear = -1;

void enqueue(int data) {
    if ((rear + 1) % max == front) {
        printf("Queue full\n");
        return;
    }

    if (front == -1) {
        front = rear = 0;
    } else {
        rear = (rear + 1) % max;
    }

    queue[rear] = data;
}

int dequeue() {
    if (front == -1) {
        return -1;
    }

    int data = queue[front];

    if (front == rear) {
        front = rear = -1;
    } else {
        front = (front + 1) % max;
    }

    return data;
}

int main() {
    int n, x, removed;

    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &x);
        enqueue(x);
    }

    removed = dequeue();

    if (removed == -1) {
        printf("Queue empty\n");
    } else {
        printf("%d\n", removed);
    }

    return 0;
}
```

Test Input

```
5
10
20
30
```

Output

```
10
```

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <ctype.h>
#include <string.h>

#define max 100

int stack[max];
int top = -1;

void push(int x) {
    if (top < max - 1) {
        stack[++top] = x;
    }
}

int pop() {
    if (top == -1) {
        return -1;
    }
    return stack[top--];
}

int evaluatePostfix(char expression[]) {
    int i = 0;

    while (expression[i] != '\0') {
        if (isdigit(expression[i])) {
            push(expression[i] - '0');
        } else {
            int op2 = pop();
            int op1 = pop();
            int result = 0;

            switch (expression[i]) {
                case '+':
                    result = op1 + op2;
                    break;
                case '-':
                    result = op1 - op2;
                    break;
                case '*':
                    result = op1 * op2;
                    break;
                case '/':
                    result = op1 / op2;
                    break;
            }

            push(result);
        }
        i++;
    }

    return pop();
}

int main() {
    char expression[max];

    scanf("%99s", expression);

    printf("%d\n", evaluatePostfix(expression));

    return 0;
}
```

Test Input

```
23+
```

Output

```
5
```

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